ABV–Indian Institute of Information Technology and Management (ABV-IIITM), Gwalior

Software Engineering Laboratory

Lab Sheet – 2

Date: 04 August 2026

Course: Software Engineering Laboratory

Experiment No.: 2

Title: Library Management System – Functional Testing

Instructor: Dr. Dheeraj Kodati

Assistant Professor

Department of Information Technology

Aim

To develop a Library Management System and validate its functionality using software testing techniques.

Problem Statement

An Excel spreadsheet containing **10–20 library book records** will be provided by the instructor.

The spreadsheet contains the following columns:

- Book ID
- Book Title
- Author
- Category
- Total Copies
- Available Copies
- Issued Copies
- Status (Empty)

The following relationship must always hold:

Total Copies = Available Copies + Issued Copies

Write a program that:

- Reads the uploaded spreadsheet.
- Validates each book record.
- Determines the current status of the book.
- Updates the **Status** column.

Status Policy

Condition	Status
Available Copies > 0	Available
Available Copies = 0	Issued Out
Invalid record	Error

Laboratory Tasks

Task 1

Download the Excel sheet uploaded by the instructor.

Task 2

Write a program to:

- Read the spreadsheet.
- Validate the book information.
- Determine the Status.
- Update the Status column.

Task 3

Execute the following test cases by adding one new book record for each case.

Test Cases

Test Case 1 – Invalid Book Record

Add a record where

- Total Copies = 10
- Available Copies = 7
- Issued Copies = 5

Since

7 + 5 = 12 ≠ 10

the record is invalid.

The program should display an appropriate error message and should not assign a status.

Test Case 2 – Book Completely Issued

Add a book with

- Total Copies = 5
- Available Copies = 0
- Issued Copies = 5

Execute the program and verify whether the generated status is **Issued Out**.

Test Case 3 – Missing Input

Add a record where any one of the following fields is left blank:

- Total Copies
- Available Copies
- Issued Copies

Execute the program and observe how missing values are handled.

Test Case 4 – Invalid Data Type

Add a record where

Available Copies = "Five"

instead of a numeric value.

Execute the program and observe the system behavior.

Record the generated error.

Test Case 5 – Negative Inventory

Add a record with

- Total Copies = 10
- Available Copies = -2
- Issued Copies = 12

The program should identify that negative inventory values are invalid and reject the record.

Instructions

- Use the uploaded Excel sheet as the input.
- Update the Status column only for valid records.
- Display meaningful error messages for invalid records.
- Record the outcome of each test case as PASS or FAIL with justification.